As briefly described in Section ?? the software for NA62 is split into three main `C++` packages. NA62MC is a GEANT4 based Monte Carlo simulation package that simulates the NA62 experiment. The MC utilises functions within GEANT4 to construct the sub-detectors of NA62 including all their materials and structures. It is possible to select a number of K decays to simulate to create signals similar to what are produced in real life at the experiment. After the simulation is complete the software saves the signals in each detector in a ROOT file for use in reconstruction or analysis. NA62Reconstruction is then used to reconstruct either the Monte Carlo data or collected data from the experiment, producing hits within the detectors and clustering them into candidates. NA62Analysis is used to analyse the data produced from the reconstruction to conduct all the physics studies that are conducted at NA62.

The na62fw was designed for use only with the NA62 experimental setup, resulting in in minimal configurability and flexibility. Within the simulation the only option for users to change was to enable/disable sub-detectors. With the experiment foreseen to run up to CERNs Long shutdown 3 (LS3), an upgraded software package was needed to allow for the development of new detectors for the future kaon experiments. The software had to be flexible and configurable to allow for the simultaneous use of the software for NA62 data taking and physics analysis, as well as the development of new experiments.

0.1 FlexibleMC

0.1.1 Description

For the implementation of a flexible framework all detector classes were refactored to inherit from a common class which contained the basic functions and variables for all detectors. This allowed for all classes for the detectors to be parsed using the same machinery. The input to the simulation was then changed from a simple text file, to a single configuration file for all the detectors. The configuration file contained any arguments required for individual detectors which are defined in Table 1. The X,Y,Z placement of the detectors and rotation were also defined in this file. An example entry in the configuration file is shown in Figure 1 where a name was given for the station ”STRAW0”, along with chamber ID, chamber type which defines the size of the straws and the chamber geometry which defines if the beam hole is symmetric or not.

```
       Name              Arguments        X Y    Z       θx θy θz
Spectrometer("STRAW0", 0, 0, 0) 0 0 183705.9 0  0  0
```

Figure 1: Example configuration for the STRAW0 in the geometry file for NA62MC.

Detector Name	Arguments
STRAW	Chamber Name: ”STRAW0”,”STRAW1”,”STRAW2”,”STRAW3” Chamber ID: 0,1,2,3 Chamber Type: 0(10mm), 1(5mm) Chamber Mode: 0(default), 1(symmetric)
Spectrometer Magnet	Magnetic field intensity
Vessel	Length InnerRadius OuterRadius Type(”BlueTube”, ”BlueTubeNoRails”, ”GreyTube”)

Table 1: Arguments for detectors in the geometry file for use within NA62MC.

When constructed inside the simulation, each detector was contained within its own separate outer volume called the ”Responsibility Region” (RR). The creation of the RR was a simple task for most detectors apart from the STRAWs and LAV which were made up of multiple stations that were not continuous, resulting in multiple responsibility regions for each station.

0.1.2 Testing: Geometry and Magnetic Fields

After the simulation was upgraded all the geometries of each detector had to be validated that they were still correct with respect to the previous version. This was due to the complex nature of refactoring the detectors with a number of volume changes and the way the detectors were constructed. The validation was completed using a virtual particle within the GEANT4 framework called a Geantino. The

Geantino is a particle which will traverse the simulation in a straight line and interact with all volumes present in the simulation but will not be impacted by magnetic fields. A large square beam of these particles was sent through the simulation and every time the particle passed across a boundary between two volumes its position was saved. Using this information it was possible to reconstruct the entire geometry of each detector in order to check the placement and geometry to mm precision to ensure that the simulation was still accurate. An example of the Geantino hits on a single view of STRAW chamber 1 is shown in Figure 2 where it is possible to see the features of the view, such as the missing straws within the center, used to produce the chamber beam hole. For the full validation a larger sample of Geantinos was used to get the required precision. However, if a discrepancy between the new version and the previous, a smaller area was selected to create a more precise picture on what was incorrect. This information was then used to modify the software to fix any geometry issues. This check was conducted for all sub-detectors.

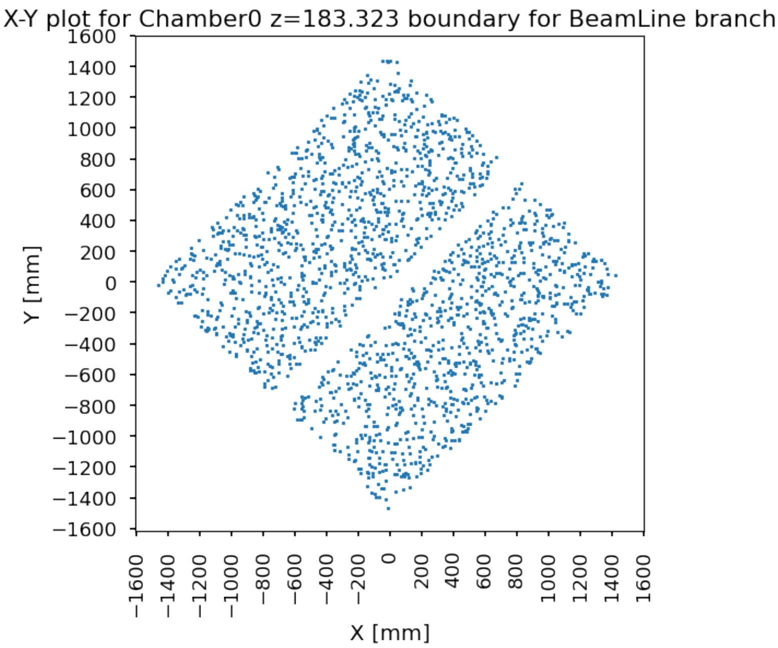


Figure 2: Geantino hits on a single view in STRAW chamber 1.

To reduce the chance of overlaps and to clean up the simulation, the size of the unused space in all sub-detectors surrounding the physical geometry was reduced. The RR for all sub-detectors was reduced to just slightly larger than the physical detector. It is possible to see this change in figure 3, which depicts the direction of the magnetic fields. As the magnetic fields are only calculated in volumes, it is possible to see the reduction in RR size around the geometry. In 3(top) the field was calculated outside of the actual physical sub-detector due to the larger RR, whereas in 3(bottom) the RR has been reduced to a few mm's bigger than the actual physical detector. As well as reducing the chance of overlaps, it would speed up the simulation due to the reduction in unnecessary calculations for the magnetic field outside of the detector geometry.

To improve the usability for users a geometry check is completed every time before the simulation is ran. Each outer detector volume is checked for any overlaps with other detectors in all three dimensions taking into account any rotations and the shape of the outer volumes. If an overlap of two sub-detectors is found the simulation does not start and an error is produced to the user describing where the overlap is and what sub-detectors are involved.

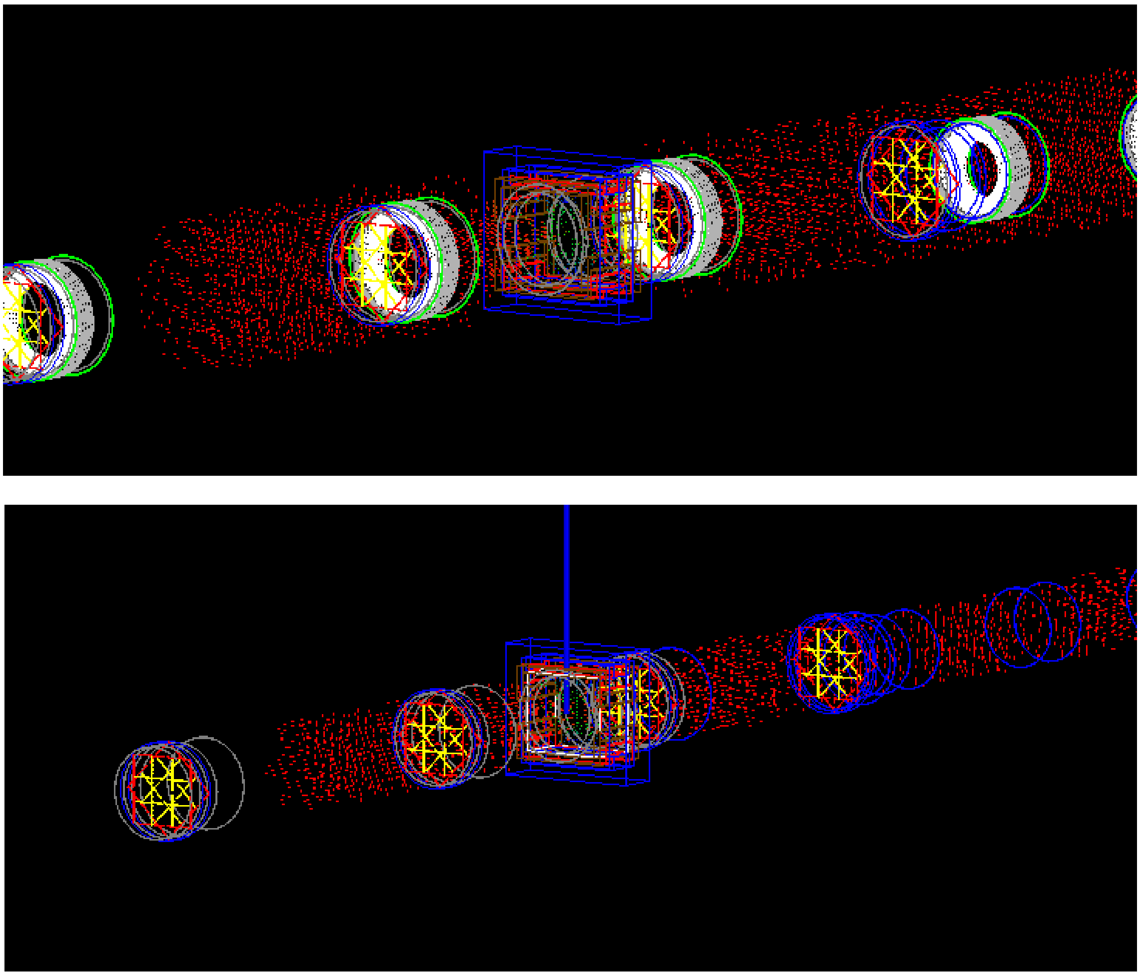


Figure 3: Visualisation of the magnetic fields in the spectrometer region from the simulation. Top, previous version with large outer volumes and bottom, new flexible version with reduced outer volumes. The small red arrow represent the direction of the magnetic field in the volumes.

The magnetic field in the simulation is based of real-world measurements of the magnetic field at NA62 from hall probes [**?**], with this information saved in the form of data files which is taken in and used to calculate the magnetic field at various points in the simulation. In NA62MC the magnetic field is split into three distinct regions; Bluetube which is the magnetic field within the decay region which is used for small corrections, fringe field which are the edges of the MNP33 magnetic field and the MNP33 field which is the main magnetic field defined in the simulation from the MNP33 dipole magnet. In the previous version of the simulation the position of the magnetic fields were written within the static code. However, in the flexible version it was possible to move the MNP33 magnet to be placed in any position via the geometry input file. This required for the magnetic field to be calculated relative to the central position of the MNP33 dipole magnet. An example of this flexibility is shown in Figure 4 where the MNP33 magnet was moved from 198 m to 163 m, which is in front of the STRAW chamber 1. The MNP33 and fringe magnetic field was recalculated from the new position at 163 m to account for the placement of the MNP33 magnet.

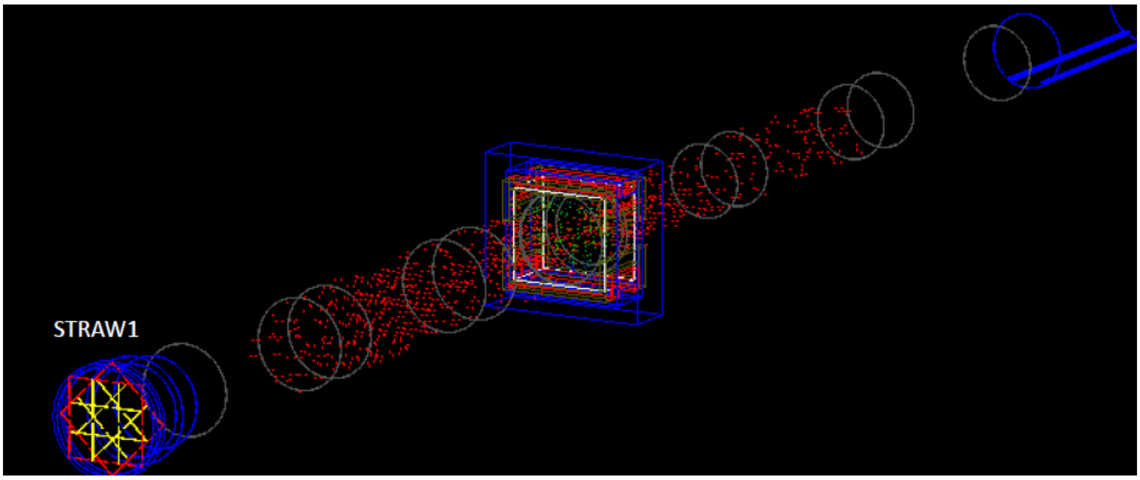


Figure 4: Visualisation of the MNP33 magnet placed in front of the STRAW chamber 1 at 163 m from the target in the simulation. Originally the magnet was placed at 198 m, which is after the STRAW chamber 2. The small red arrow represent the direction of the magnetic field in the volumes.

Similar to the changes in the sub-detector construction, the modification of the magnetic fields required validation. This was done using similar machinery to the geometry test completed previously but with different GEANT particle, the charged Geantino. Similar to a normal Geantino, where the particle traverses all volumes within its path but interacts with magnetic fields. As the charged Geantino enters a magnetic field it is deflected as if it were a positively charged particle. This test ensured that the same tracks of the charged Geantinos between the previous version and the new flexible version were being produced within the magnetic fields. An example of the tracks taken by the charged Geantinos within the simulation is shown in Figure 5 where it is possible to see the displacement of the particles within the magnetic field regions. The exact XY positions of the hits would then be validated against a sample of charged Geantinos with the exact same characteristics and starting parameters produced with the previous version of the software.

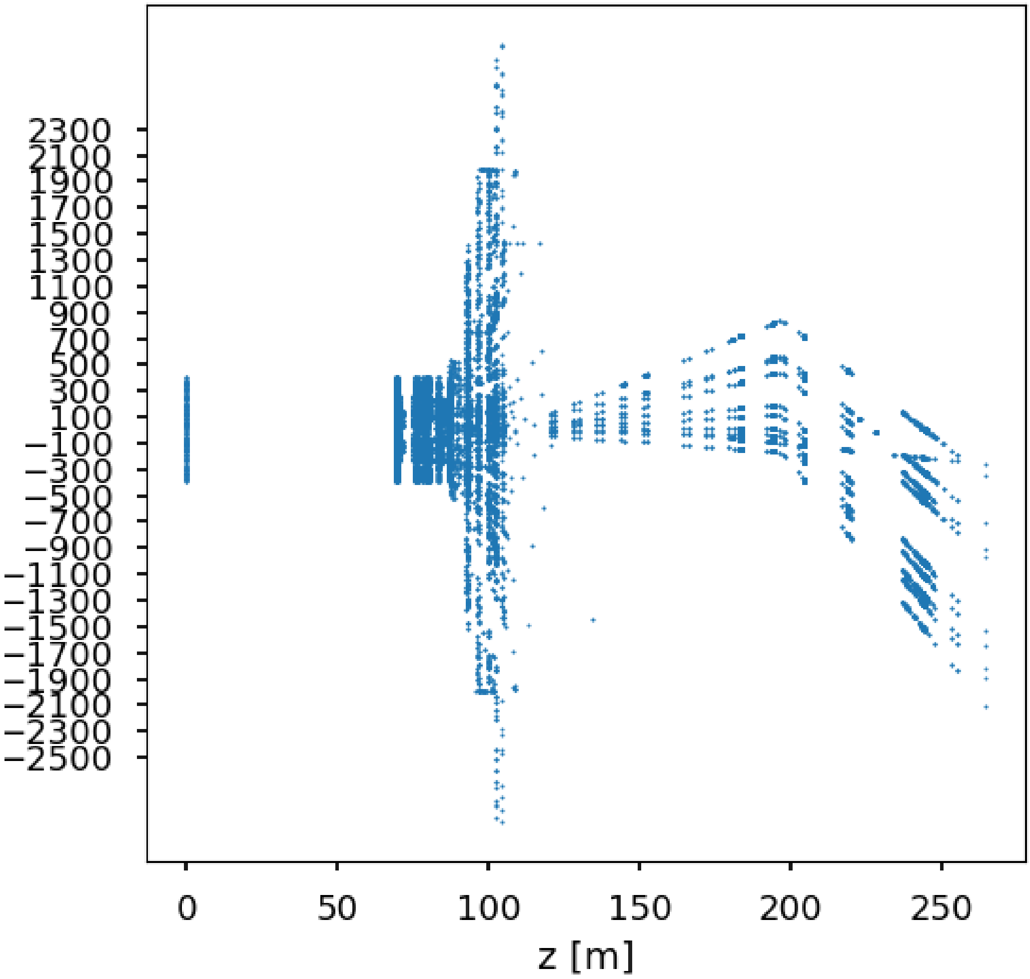


Figure 5: Charged Geantino hits on all detectors within the simulation.

With minimal modification it was also possible to add the functionality to add multiple magnets in the simulation with their magnet fields producing a combined field. This was not implemented into the framework as its use was not foreseen but the machinery was developed and is ready for uses when required.

After the geometry and magnetic field tests were completed, full test productions of $K^+ \to \pi^+\pi 0$, $K^+ \to \pi^+\pi^-\pi^+$ and $K^+ \to \mu^+\bar{\nu}$ were produced to test the validity of the existing NA62 setup along with the entire software chain (MC-¿Reconstruction-¿Analysis).

0.1.3 Testing: Symmetric NA62

It was also required to test the entire software chain using the new features of the flexible framework to ensure that when it was required and used in the future there were no issues. It was proposed to conduct a small study to investigate the feasibility of a symmetric NA62.

As described previously in Chapter **??** the current NA62 detector setup has been optimised for the search of $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ with the current experiement set to run until CERNS LS3. After LS3 it was foreseen that there would be the possiblilty of both charged and neutral kaon experiemnts located at CERN requiring a redesign of the NA62 detector setup. In the current NA62 setup, the beam is deflected in the positive x-direction by the TRIM5 magnet, which is located between GTK stations 2 and 3. The TRIM5 has a strength of $-0.3\,Tm$, which equates to a 90 MeV kick, deflecting the beam to a maximum deflection of approximately +114 mm at straw chamber 2. After chamber 2, the MNP33 dipole magnet deflects the beam in the negative x direction with a momentum kick of 270 MeV. This requires the beam holes for the four Straw stations to be offset from the centre axis as shown in Figure 6 to allow for the undecayed charged beam to through the spectrometer. The mean beam centre of a charged Kaon at $75\,GeV$ is shown by the red line starting from straw chamber 1 all the way to the LKr accounting for the interactions with the TRIM5 and the MNP33 magnets.

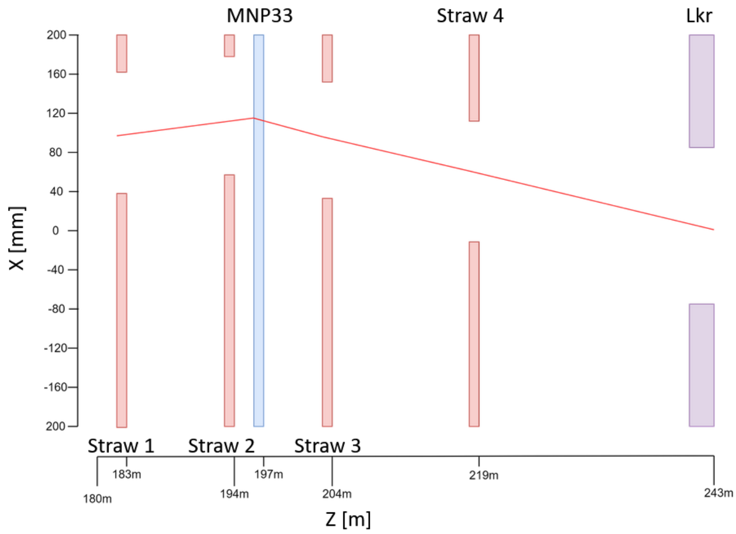


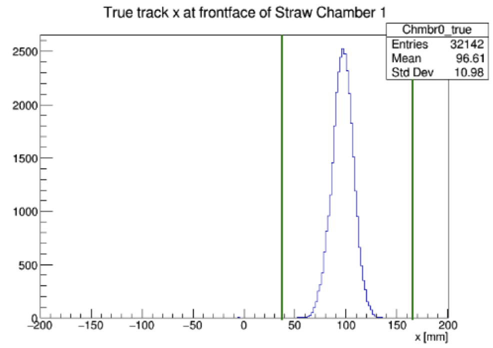
Figure 6: Diagram showing the path of the charged kaon beam, the straw chamber beam holes and the LKr beam pipe

The exact positions of the NA62 straw chambers and the beam holes are defined in Table 2.

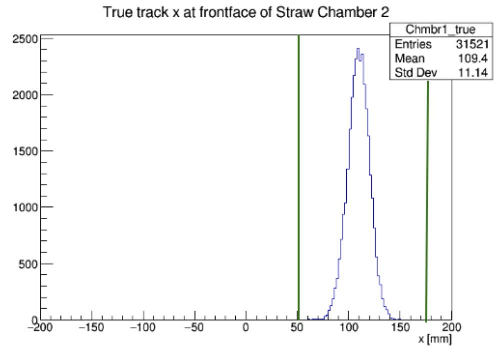
Straw Chamber	Z BackFace [m]	X Hole Displacement [mm]
1	183.7	101.2
2	194.26	114.4
3	204.64	92.4
4	219.07	52.8

Table 2: Straw chamber beam hole properties for NA62 detector setup

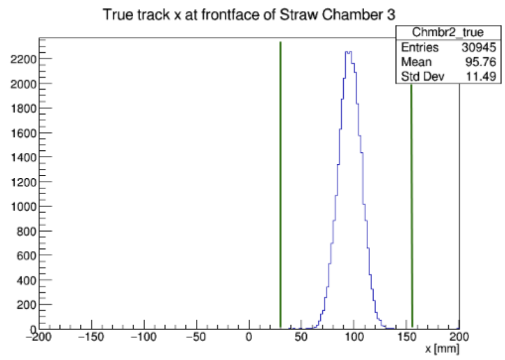
The beam profile at each Straw station is shown in Figure 7.



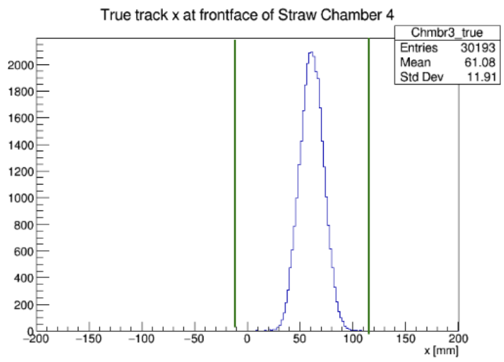
(a) Straw chamber 1



(b) Straw chamber 2



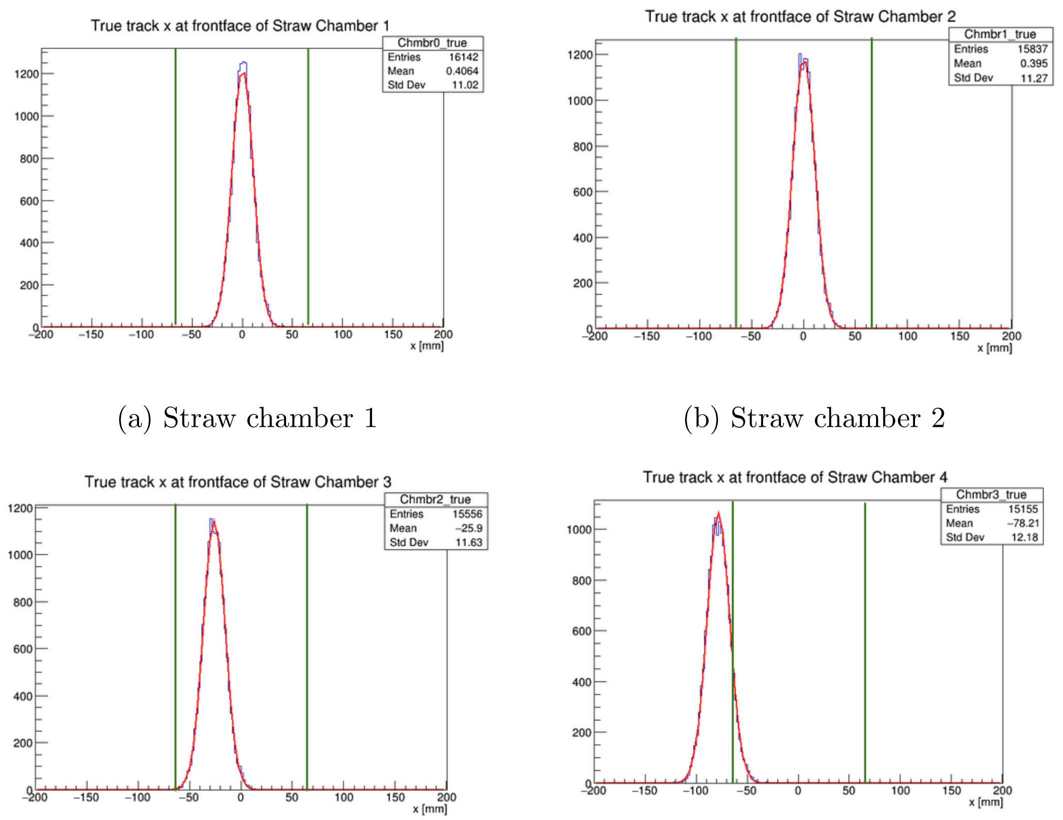
(c) Straw chamber 3



(d) Straw chamber 4

Figure 7: Plots of current NA62 beam coordinates at each straw chamber with the beam hole walls noted by the green vertical lines

To accommodate a neutral beam, the beam holes in the Straw stations would have to become centred at 0 in the X dimension, creating a symmetric detector design. However using the current NA62 beam setup it is clear the hole centred at 0 would be incompatible as from Figure 7 the beam would impact the detector. To fix this issue the TRIM5 magnet would be removed, allowing the beam to traverse along the z-axis until the interaction with the MNP33 magnetic field allowing the beam to pass through Straw chambers 1 and 2 but resulted in the beam impacting chamber 4 as shown in Figure 8.



(a) Straw chamber 1

(b) Straw chamber 2

(c) Straw chamber 3

(d) Straw chamber 4

Figure 8: Plots of symmetric straw chambers with current NA62 beam parameters

In order to fit the beam within the last straw chamber three paramters from the NA62 could be modified:

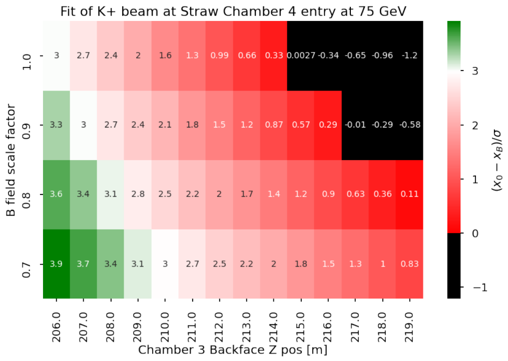
1. Beam momentum (NA62 Nominal 75 GeV)
2. Chamber 4 positioning (NA62 Nominal back-face at 219.07 m)
3. MNP33 magnet field intensity (NA62 Nominal at 0.9 Tm, which equates to a 270 MeV momentum kick)

To qunatify if the beam fitted within the beam hole of the detector a saftey parameter was defined:

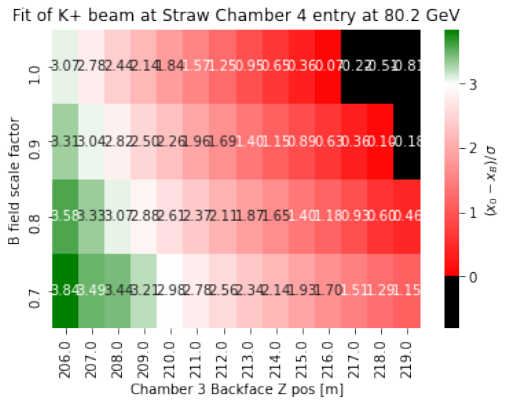
$$\delta = \frac{x_0 - x_d}{\sigma} \tag{0.1}$$

Where x_0 is the mean of the fitted Gaussian of the beam at the detector plane, x_B is the x position of the beam hole wall (which was -63.8mm for the symmetric straw chambers), and σ is the sigma of the plotted Gaussian of the beam at the detector plane. The beam is defined to have fitted in the beam hole if $\delta \geq 3$.

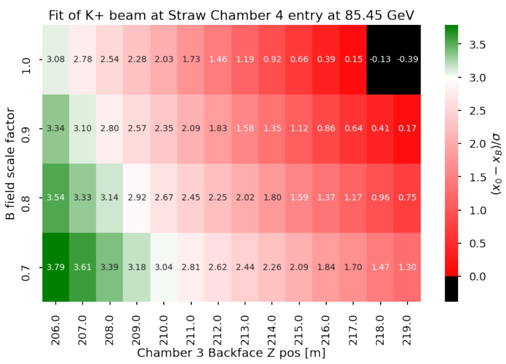
The results of modifying the three parameters and their impact on the beam profile at straw chamber 4 is shown in Figure 9. Comparing results first for different momentum there is no clear change between 75 to 85 GeV due to the size of the beam increases as momentum increases counteracting the reduction in the deflected produced by the MNP33 magnet (MAYBE PUT BEAM SIGMA IN APPENDIX). Taking into account tht switching the beam momentum would cause other issues to the beam infrastructure it was decided to only continue with a 75 GeV beam (NA62 Nominal).



(a) 75 GeV beam momentum



(b) 80.2 GeV beam momentum



(c) 85 GeV beam momentum

Figure 9: Plots of δ for different values of straw chamber 4 z position, MNP33 magnetic field and beam momentum

Focusing on Figure 9(a), only a small number of combinations produce a satisfactory safety factor. The optimat solution was with the MNP33 B field factored down by 0.7 and the straw station 4 moved to 206 m, however this would place chamber 4 extreamly close to chamber 3 which would impact the reconstruction resolution and would become apparent below. It was expected that a chamber position of

$> 209\,m$ would produce a acceptable resolution leaving few combinations available, other than ones at 0.7 MNP33 field scale. The setup with chamber 4 at 209 m was chosen to ensure that the beam definitely fitted within the beam hole.

The next phase was to ensure that the beam would safely pass through the beam hole through the LKr detector. This was imperative as unlike all most all other detectors which would be simple to move or slightly their positions, due to its size and position in the beam line it would make it difficult if not impossible to move. The Lkr as described in Chapter **??** is located 243 m downstream from the target and is slightly offset in the x dimension by 4.9 mm. It has a beam pipe which passes through its centre with a radius of 80 mm and a thickness of 3 mm. As the MNP33 magnet deflects the beam in the negative x direction, the wall of the beam pipe of interest was located at -75.1 mm. It was noted that instead of moving the STRAW chamber 4 upstream by 10 m to 209 m, it would have be more beneficial to move the other chambers and the MNP33 magnet downstream towards chamber 4 by 10m instead, as shown in the Figure 10. However this produced results with the beam impacting the LKr detector at -84.49 mm. Not many options were left as the entire spectrometer could not be moved further downstream due the placement of the RICH, the distance between chamber 3 and 4 was already at its minimum and it was assumed that the distance between the MNP33 magnet and chamber 3 had previously been optimised.

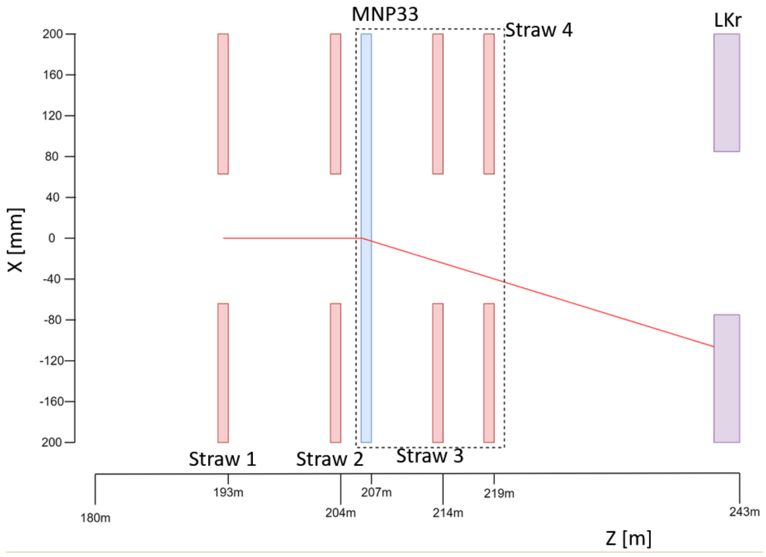


Figure 10: Diagram showing the updated setup with straw chambers 1,2,3 and the MNP33 magnet moving 10 m downstream

A solution was found that included utilising the previously removed TRIM5 magnet, displacing the charged kaon beam in the positive x direction before the MNP33 magnet. To allow for the biggest momentum kick possible straw chambers 1 and 2 were move back to their original positions, resulting in a modified setup shown in Figure 11.

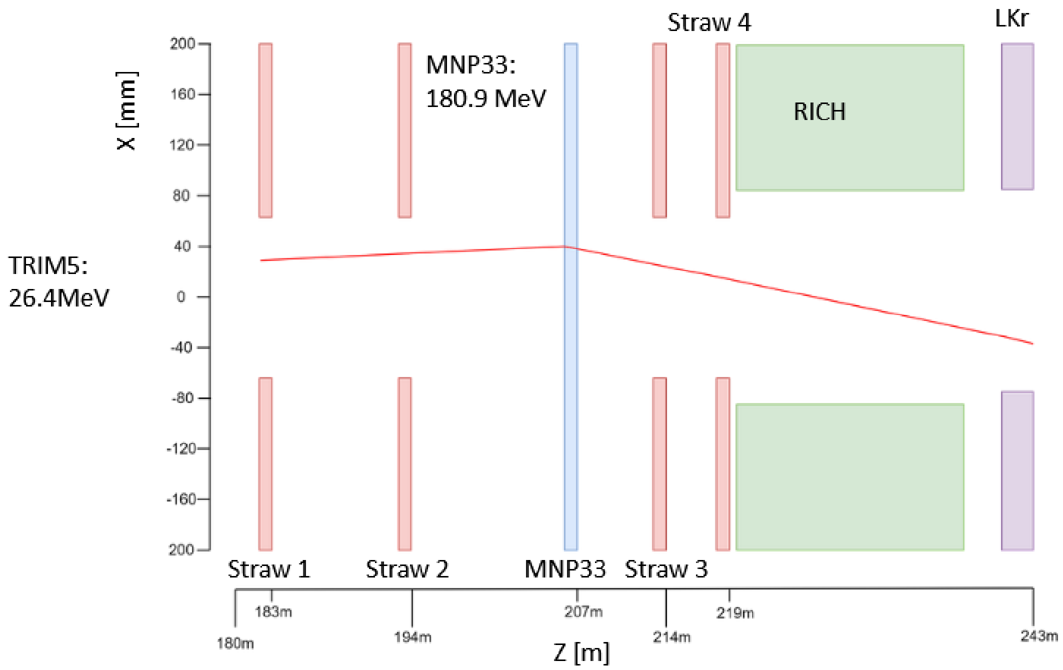
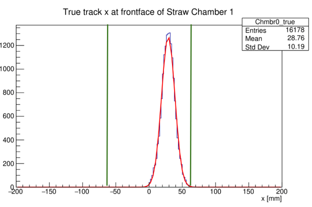
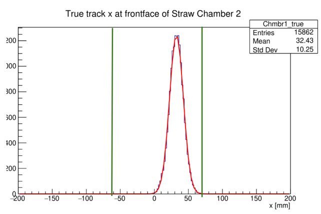


Figure 11: Diagram showing the proposed NA62 setup

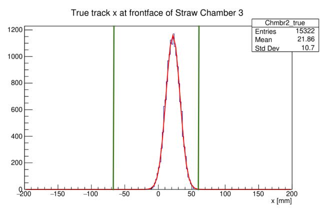
The solution which provided the best beam placement in all straw chambers and the LKr and shown in Figure 12 was found when TRIM5 produced a -26.6 MeV kick (originally -90 MeV) and the MNP33 field scale was set at 0.67, producing a kick of 180.9 MeV (originally 270 MeV).



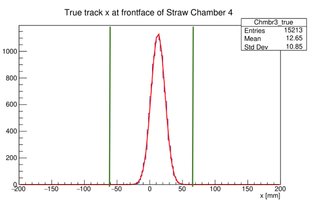
(a) Straw chamber 1



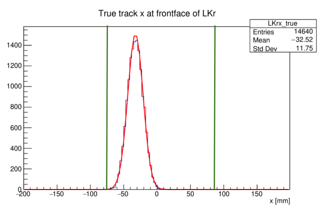
(b) Straw chamber 2



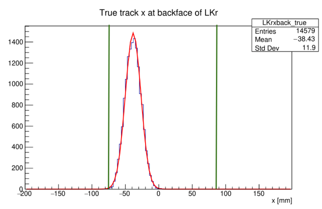
(c) Straw chamber 3



(d) Straw chamber 4



(e) LKr front-face



(f) LKr back-face

Figure 12: Plots of the beam profile in each of the straw chambers and the LKr front and back faces for the proposed symmetric setup

The overall safety margins for each Straw station and the LKr is provided in Table 3 with the following detector setup:

- MNP33 new z back-face position = 207.645 m (NA62 Nominal 197.645 m)
- MNP33 new field momentum kick = 180.9 MeV (NA62 Nominal 270 MeV)
- Chamber 3 new z back-face position = 214.645 m (NA62 Nominal 204.65 m)
- TRIM5 new field momentum kick = -26.6 MeV (NA62 Nominal -90 MeV)

Detector	Saftey Margin	Number of beam particle outside of beam pipe/hole %
Chamber 1	3.46	0.035
Chamber 2	3.09	0.11
Chamber 3	3.99	0.025
Chamber 4	4.81	0.03
LKr Front	3.70	0.045
LKr Back	3.16	0.115

Table 3: Results showing the safety margin and number of original beam particles hit outside of the beam hole/pipe boundary for each chamber and the LKr

For this study only the required detectors to produce a final result were modified, however if this were to be implemented in the future other detectors would have to be modified. For example the GTk would have to be flexible enough for it to be removed from the beamline for the neutral mode. Downstream from the RICH would require minimal modification as shown in Figure 13, with the main issue being the beam pipe located in the downstream section of the HAC. It is acceptable for regular operation, but when the polarities of MNP33 and TRIM5 are reversed, the beam would fall outside the beam pipe. A much wider beam pipe, symmetric along the z-axis would have to be implemented.

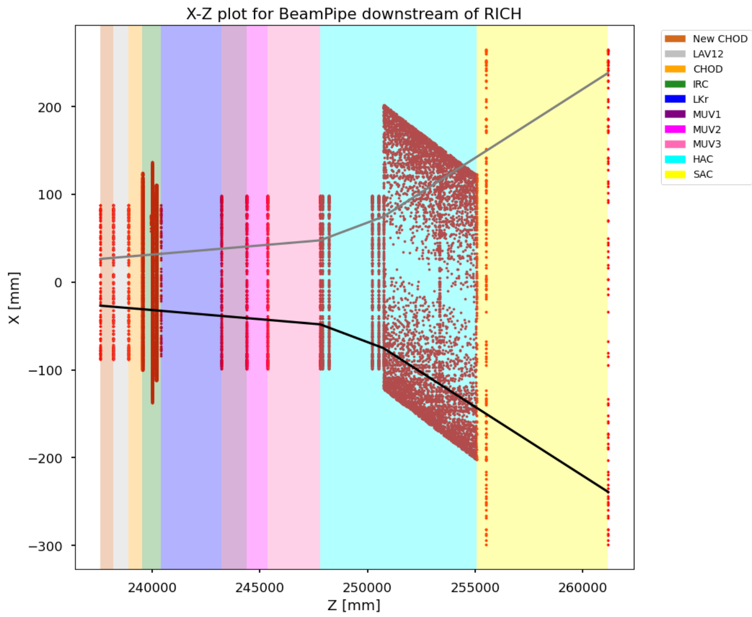
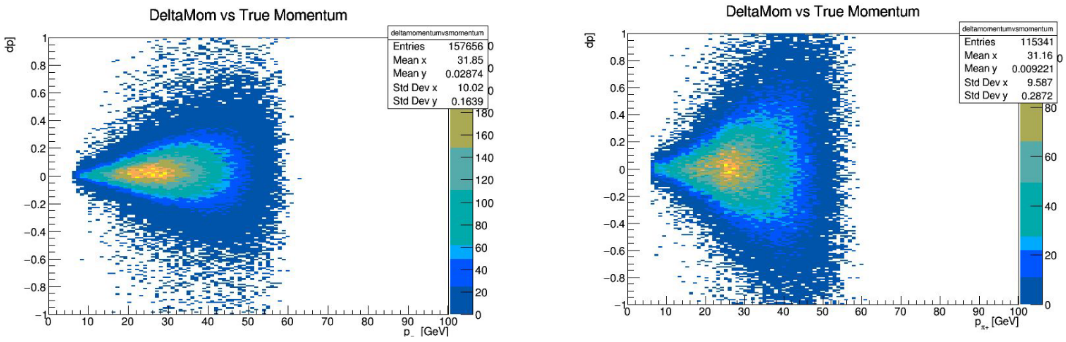


Figure 13: Plot produced from Geantino study of detectors beam pipes downstream of the RICH. The black line shows the centre of the proposed beam, and the grey shows the proposed beam with the magnet polarities reversed

To determine if the symmetric setup defined above was viable the resolution of the reconstructed momentum of charged particles and then the resolution of the squared missing mass would have to be quantified. The squared missing mass resolution is critical as this will affect the size of the signal regions for $K^+ \to \pi^+\nu\bar{\nu}$ discussed in Chapter **??**.

A NA62 Monte Carlo sample of $K^+ \to \pi^+\pi^0$ was produced with the symmetric setup and reconstructed with NA62 Reconstruction. A generic selection for K2pi was used which uses a small number of detectors; GTK, Spectrometer, LKr, LAV, SAC and IRC. The selection involved selecting the reconstructed π^+ track in the spectrometer, and the two photons in the LKr which produced from the π^0 decay. The GTK is used for the kaon momentum calculations, and the LAV, IRC and SAC are used for photon veto.

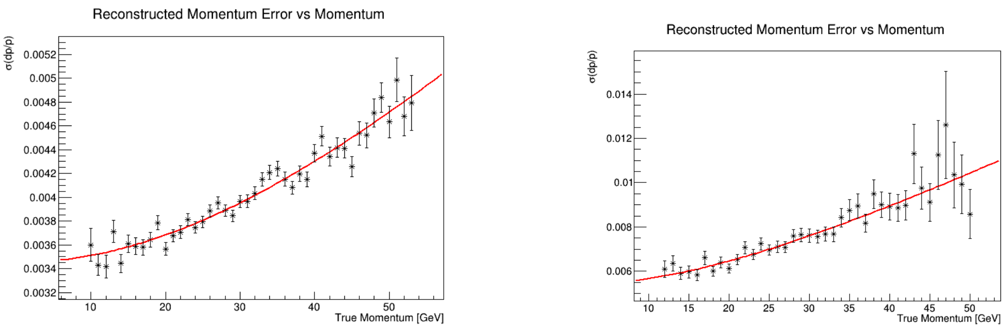
The momentum resolution was computed using the reconstructed π^+ momentum compared against the true momentum from the MC. The difference between the reconstructed and true momentum was then plotted against the true momentum for both the NA62 and the symmetric setup, shown in Figure 14.



(a) NA62 setup (b) NA62 symmetric setup

Figure 14: Difference between reconstructed and true pion momentum against true pion momentum from $K \rightarrow \pi^{+i} 0 decay$

For each true pion momentum bin a Gaussian fit was completed on the y distribution and the sigma of the fit is shown in Figure 15 for both NA62 and the symetric setup.



(a) NA62 setup (b) NA62 symmetric setup

Figure 15: Sigma of fitted Gaussian of the reconstructed and true pion momentum against true pion momentum from $K \rightarrow \pi^{+i} 0 decay$

The resolution was computed by fitting the distributions in Figure 15, with the current NA62 setup resolution calculated as,

$$\frac{\sigma_p}{p} = 0.345(\pm 0.002)\% \oplus 0.00638(\pm 0.00011)\% p \tag{0.2}$$

which is worse than the published NA62 resolution results [?] $\frac{\sigma_p}{p} = 0.3\% \oplus 0.005\% p$ due to the much looser cuts on the analysis selection used in this study. The symmetric setup momentum reslution was computed as,

$$\frac{\sigma_p}{p} = 0.539(\pm 0.0130)\% \oplus 0.0179(\pm 0.0006)\% p \tag{0.3}$$

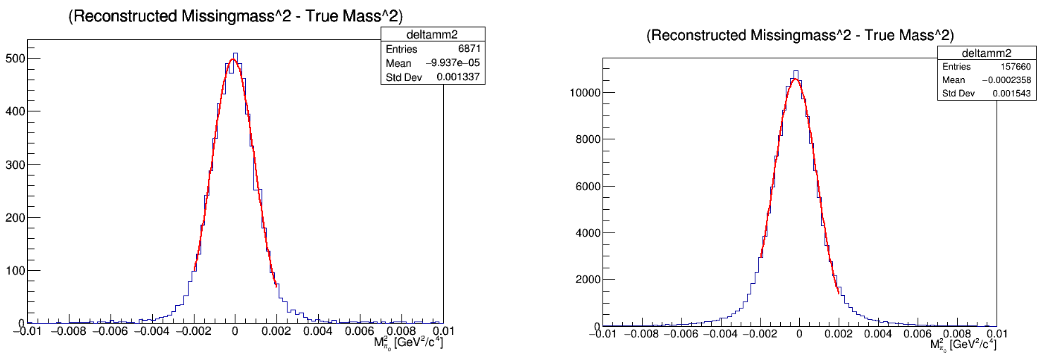
The momentum resolution due to the multiple scattering, which is the dominant factor has degraded by a factor of 1.6 with the resolution due to the position measurement degraded by a factor of 2.8. It is evident that the change in the setup has caused a degradation in the momentum resolution, however this was the first iteration of the design and was only optimised for geometrical acceptance of the beam. In future is would of been beneficial to optimise the setup for the resolution. The next step was to calculate the resolution on the squared missing mass. This is computed from the missing mass of the vertex which was the difference between the Kaon energy and the π^+, and this was compared against the π^0 mass. This was done for the NA62 and the symmetric setup and is shown in Figure 16. The resolution was taken from the fitted Gaussian for NA62,

$$\sigma(m^2_{miss}) = 1.097(\pm 0.004) \times 10^{-3} GeV^2 \tag{0.4}$$

and the symmetric setup,

$$\sigma(m^2_{miss}) = (1.490 \pm 0.007) \times 10^{-3} GeV^2 \tag{0.5}$$

resulting in a degradation of the squared missing mass resolution by a factor of 1.35 for the symmetric setup compared to the current NA62 setup.



(a) NA62 setup

(b) NA62 symmetric setup

Figure 16: Difference of Squared missing mass and the squared π^0 mass from $K \rightarrow \pi^{+i} 0 decay$

The degradation suggests that it could be possible to construct a symmetric NA62, however more optimisation would hve to be done to get as close as possible to NA62 resolutions. However this study has proved that the flexible framework developed and described above is able to accepted new detector setups and complete full feasibility studies all the way up to the analysis level. Showing that the framework is ready for new kaon experiments which will be explored in Chapter **??**.

0.2 SlimMC